

A counterexample to the equality clause of conjecture 00000000240

C0ldSmile

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Abstract

The product of a segment and a three-dimensional cross-polytope is a four-dimensional origin-symmetric convex body with Mahler volume $32/3 = 4^4/4!$. It has twelve extreme points, whereas a four-dimensional cube and cross-polytope have sixteen and eight. Thus it attains the stated bound without belonging to either claimed equality class. The accompanying Lean project proves the actual body, polar, Lebesgue volumes and obstruction to every linear image. The conclusion concerns the equality classification in the repository's statement; it makes no claim to resolve the general Mahler lower-bound problem.

1 Statement and scope

For a positive integer n and an origin-symmetric convex body $S \subset \mathbb{R}^n$, let

$$S^\circ = \{y \in \mathbb{R}^n : \langle x, y \rangle \leq 1 \text{ for every } x \in S\}, \quad M(S) = \text{vol}_n(S) \text{vol}_n(S^\circ).$$

A convex body here is compact, convex, and has nonempty interior in the ambient space. Volume is ordinary Lebesgue volume, with unit coordinate boxes having volume one. Put

$$C_n = [-1, 1]^n, \quad B_1^n = \left\{x \in \mathbb{R}^n : \sum_{i=1}^n |x_i| \leq 1\right\} = \text{conv}\{\pm e_1, \dots, \pm e_n\}.$$

The source asserts $M(S) \geq 4^n/n!$ and gives only linear images of C_n or B_1^n as equality cases. The English text asserts the necessary direction of this classification; the Chinese text states it as an if-and-only-if. We refute their common necessary direction. A false four-dimensional instance refutes either universal assertion, including its conjunction with the lower bound. The source's parenthetical reference to dimensions at most three does not restrict its general claim.

2 The body and its polar

Use coordinates (s, x_1, x_2, x_3) and define

$$K = [-1, 1] \times B_1^3 = \{(s, x) : |s| \leq 1, |x_1| + |x_2| + |x_3| \leq 1\}.$$

Both factors are compact, convex and symmetric about zero. The open set

$$\{(s, x) : |s| < 1, |x_1| + |x_2| + |x_3| < 1\}$$

contains zero and lies in K . Consequently K is a genuine four-dimensional origin-symmetric convex body.

Proposition 1. *The actual polar of K is*

$$P = \{(t, y) : |t| + |y_i| \leq 1 \text{ for } i = 1, 2, 3\} = \{(t, y) : |t| + \max_i |y_i| \leq 1\}.$$

Proof. If $(t, y) \in K^\circ$, test its defining inequality against $(\text{sgn}(t), \text{sgn}(y_i)e_i) \in K$, choosing either sign when a coordinate is zero. This gives $|t| + |y_i| \leq 1$ for every i .

Conversely, suppose these inequalities hold. In particular $|t| \leq 1$. For $(s, x) \in K$,

$$\begin{aligned} st + \sum_i x_i y_i &\leq |s||t| + \sum_i |x_i||y_i| \\ &\leq |t| + (1 - |t|) \sum_i |x_i| \leq 1. \end{aligned}$$

Thus $(t, y) \in K^\circ$. □

3 Exact Lebesgue volumes

The three-dimensional cross-polytope consists of eight orthant simplices, each of volume

$$\int_0^1 \int_0^{1-u} \int_0^{1-u-v} dw dv du = \frac{1}{6}.$$

Intersections along coordinate planes have zero three-dimensional measure. Hence $\text{vol}_3(B_1^3) = 4/3$ and the product formula gives

$$\text{vol}_4(K) = 2 \cdot \frac{4}{3} = \frac{8}{3}.$$

The Lean proof obtains the same actual cross-polytope volume from Mathlib's proved formula for the volume of ℓ_p balls, at $p = 1$ and dimension three, with the Gamma factors evaluated exactly.

At a fixed t with $|t| \leq 1$, the section of P is the coordinate cube

$$[-(1 - |t|), 1 - |t|]^3,$$

of volume $8(1 - |t|)^3$. If $|t| > 1$, that section is empty. The set P is closed and therefore measurable. The product-measure formula, followed by an ordinary integral on a compact interval, yields

$$\text{vol}_4(P) = \int_{-1}^1 8(1 - |t|)^3 dt = 16 \int_0^1 (1 - t)^3 dt = 4.$$

In particular both volume factors are finite and positive. Since $P = K^\circ$,

$$M(K) = \frac{8}{3} \cdot 4 = \frac{32}{3} = \frac{4^4}{4!}.$$

4 Excluding both classes of linear images

The extreme points of B_1^n are exactly the $2n$ signed coordinate vectors. Here is a direct justification. The inequality defining B_1^n is convex. Every $x \in B_1^n$ is the convex combination

$$x = \sum_i |x_i| (\text{sgn}(x_i) e_i) + \left(1 - \sum_i |x_i|\right) 0,$$

and zero itself is the midpoint of any opposite pair of coordinate vectors. Thus B_1^n is their convex hull. Each signed coordinate vector is extreme: a convex combination attaining its coordinate value 1 or -1 forces that value in both summands, and the ℓ_1 bound forces their remaining coordinates to vanish. A finite convex hull has no other extreme points.

The interval $[-1, 1]$ has precisely the extreme points -1 and 1 . Extreme points of a Cartesian product are the products of extreme points. Consequently

$$|\text{ext } K| = 2 \cdot 6 = 12, \quad |\text{ext } C_4| = 2^4 = 16, \quad |\text{ext } B_1^4| = 2 \cdot 4 = 8.$$

For the cube, the product rule equivalently says that every coordinate of an extreme point must be an endpoint of its interval.

Every invertible linear map carries extreme points bijectively onto the extreme points of the image. Thus K is neither an invertible linear image of C_4 nor of B_1^4 . Allowing arbitrary linear maps does not enlarge the possible image class here: if $T(S) = K$, then the linear subspace $\text{range } T$ contains a set with nonempty interior. It must be all of $\mathbb{R}^4$. A surjective endomorphism of a finite-dimensional space is invertible. This also rules out every singular map.

The equality and the two non-image conclusions contradict the necessary equality condition asserted in both language versions. Therefore conjecture 00000000240, with its stated equality classification, is false.

5 Formalization and reproducible checks

The Lean project uses Lean 4.19.0 and Mathlib v4.19.0, with the exact dependency revisions recorded in its manifest. Its ambient space is $E = \mathbb{R} \times (\text{Fin } 3 \rightarrow \mathbb{R})$, with ordinary product Lebesgue measure. The four coordinates are indexed by `Option (Fin 3)`: the distinguished coordinate is the first factor, and the other three are the second factor. The code proves the ambient dimension, expands the coordinate map and Euclidean dot product, and identifies the transported cube and cross-polytope with the inequalities used above. The coordinate equivalence is also proved to preserve volume.

Module	Mathematical content
Geometry240	Convex-hull characterization, all extreme points, and exact cardinalities for cubes and cross-polytopes.
Objects240	Concrete body and polar, compactness, convexity, interior and symmetry, dimension, coordinates, and exclusion of all linear images.
Volume240	Actual Lebesgue volumes of the cross-polytope, body and polar, including the measurable sections and evaluated integral.
Conjecture240	The explicit counterexample and negations of the four-dimensional English and Chinese assertions.

The principal theorem `Mahler240.counterexample` includes the body’s admissibility, dimension, both real volumes, equality with $4^4/4!$, and both non-image conclusions. The theorems `conjecture_00000000240_english_false` and `conjecture_00000000240_chinese_false` negate the respective dimension-four conjunctions. This specialization suffices to refute the universal conjecture. The finite volume identities justify the real-valued measure conversions; numerical constants do not stand in for geometric objects or their measures.

The independent script `reproduce.py` uses rational arithmetic to enumerate supporting facets, verify primal-polar vertex duality, construct the complete face lattices, and sum determinant volumes of barycentric simplices. Its separate section calculations agree with these volumes. Reference cube and cross-polytope volumes, and a nonsingular rational shear, check the normalization. The script reports

	vertices	facets	volume
K	12	10	$8/3$
K°	10	12	4

These computations corroborate the formal proof. The Lean theorem does not assume their output.

Reproduction instructions and exact verification records accompany the submission. The required validation covers a fresh project build, warnings-as-errors replay of every source module, the full types and axiom dependencies of authored and generated declarations, execution of all auxiliary checks, and inspection of every page of the matching PDF. Independent local verification is distinct from acceptance by repository maintainers.